

Matrix functions and stability

# Uniform polynomial bounds for products at joint spectral radius one

**Difficulty:** challenging **Importance:** interesting to the community

**Status:** Partially resolved

**Rating rationale:** Challenging because the constant must be uniform over all families of a given dimension; community impact spans transient growth and stability analysis.

## Context and notation

Let  $\mathcal{H}_d$  denote the nonempty compact subsets of  $\mathbb{C}^{d \times d}$ .

The joint spectral radius of a nonempty compact set  $\mathcal{M} \subset \mathbb{C}^{d \times d}$  is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written  $\rho(A)$ .

## Problem statement

Does each  $d \geq 1$  admit  $\Theta_d > 0$  such that every  $\mathcal{M} \in \mathcal{H}_d$  with  $\hat{\rho}(\mathcal{M}) = 1$  satisfies

$$\|A_k \cdots A_1\|_2 \leq \Theta_d (Lk)^{d-1}, \quad L = \max_{A \in \mathcal{M}} \|A\|_2,$$

for all  $k \geq 1$  and all  $A_1, \dots, A_k \in \mathcal{M}$ ?

## Reference and status evidence

Epperlein and Wirth, [The joint spectral radius is pointwise Hölder continuous](#), §2, Conjecture 3 (L3). Lemma 27 proves dimension two. The constant above must be independent of the family.

## Common follow-up screen for MF-05–MF-07

Searches combining “joint spectral radius” with “local Hölder”, “Lipschitz lower”, “trajectory bounds”, the authors’ names, and 2025/2026 found no later resolution. These searches supplement the explicit 2025 conjectures; they do not establish exhaustiveness. The three entries are separately named assertions in the source, not a count of dimensional cases.

## Audit — 2026-09-10

Rechecked [Conjecture 3 \(L3\)](#) and [Lemma 27](#): dimension two is proved, but higher-dimensional uniform constants remain conjectural. Searches for trajectory bounds and subsequent Epperlein–Wirth work found no resolution. Family-dependent polynomial estimates do not establish the displayed uniform statement.